

Multi-agent clinical consultation converges on consensus faster than on correctness

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Abstract

Multi-agent “consultation” has become the default design pattern in medical language-model systems, on the intuition that a panel of specialists diagnoses better than one physician. We test that intuition at the scale at which it has been tested for general-domain agents, and in the one regime those studies cannot reach. Medical multiple-choice reasoning uses no tools, so the tool-coordination trade-off that dominates the general-domain scaling law vanishes identically, isolating coordination itself; and expert-level medical items drive single-agent accuracy down to 15%, far below anything in existing agentic benchmarks. Across 180 multi-agent configurations and 63,499 episodes — three capability tiers, three medical benchmarks, five canonical architectures, panel sizes $N \in \{1, 3, 5, 7, 9\}$, budget-matched single-model controls throughout — three results follow.

Collaboration is bounded by a *window* of task difficulty, not a ceiling. Configurations whose single-doctor baseline falls between 25% and 50% gain +2.48 pp; those outside lose 0.31 pp (Welch $t = 8.42$, $p < 10^{-6}$). The separation is two-sided and without overlap: all 20 configurations that significantly beat the single doctor lie inside the window, all 10 that significantly lose to it lie outside. The 45% capability ceiling reported for general-domain agents is reproduced as the window’s upper edge; its lower edge is visible only at medical difficulty.

Panel size is not the design variable. It carries no predictive value whatsoever (cross-validated $R^2 = -0.006$ alone), gains saturate at three agents, and what distinguishes the best architecture from the worst is whether a competent agent audits the panel before it commits. Specialist capability matters $4.1\times$ more than the orchestrator’s, and an orchestrator too weak to audit produces the largest degradation in the study (-13.6 pp, $p = 10^{-5}$) by overriding a correct majority.

Consultation degrades the signal a clinician would use to catch its errors. Peer discussion raises unanimity from 66% to 98% while leaving the error rate given unanimity at 50%; it raises confidence on wrong answers more than on right ones, compressing the confidence gap that separates them by 45%. Accuracy gains of 6–8 pp inside the window are real, but the panel output a clinician would most readily act on — unanimous and confident — has been made the least diagnostic of its own reliability.

1 Introduction

Ask three AI specialists a hard clinical question and let them confer, and they will agree with one another 98% of the time. They will also be wrong half the time they agree. Those two facts are the subject of this paper.

Multi-agent “consultation” has become the default design pattern in medical language-model systems. MedAgents [1], MDAgents [2], MAC [3] and TeamMedAgents [4] all instantiate the same clinical intuition — a panel of specialists should diagnose better than one generalist, because a tumour board diagnoses better than one physician — and the pattern is being deployed faster than it is being measured.

The most complete measurement to date is not in medicine. Kim et al. [5] evaluated 180 controlled configurations across five canonical architectures, three model families and six agentic benchmarks under matched token budgets, and derived a quantitative scaling principle for agent systems. Three findings bear directly on clinical deployment. A *capability ceiling* ($\beta = -0.408$, $p < 0.001$): once a single agent exceeds roughly 45% accuracy, additional agents yield negative returns. A *tool-coordination trade-off* ($\beta = -0.330$, $p < 0.001$): tool-heavy tasks pay an overhead that erodes any collaborative gain. And *architecture-dependent error amplification*: independent panels amplify individual errors $17.2\times$ relative to a single agent, while centralized coordination contains them to $4.4\times$ through a validation bottleneck at the orchestrator.

Medicine was excluded from that study, and the exclusion is more interesting than it looks. The 45% threshold sits exactly where expert-level medical question answering lives: frontier models score 38–56% on MedXpertQA [6] while exceeding 90% on MedQA [7], so a single medical benchmark suite straddles the threshold from both sides. And the tool-coordination trade-off, the strongest confound in the general-domain result, is absent by construction from medical multiple-choice reasoning: the task uses *no tools at all*. Medicine is therefore not merely an untested domain but the $T = 0$ limiting case of the scaling principle — the one slice of the design space where coordination can be measured with the tool-overhead term removed, and the one regime hard enough to reveal what happens below the threshold rather than above it.

We run that experiment at matched scale: 180 multi-agent configurations, 63,499 episodes, three capability tiers, three medical benchmarks, five architectures, panel sizes $N \in \{1, 3, 5, 7, 9\}$, and a budget-matched single-model control at every cell.

Three results follow, and the third is the one a clinician should care about. First, the capability ceiling transfers, but as one edge of a two-sided *window*: collaboration pays when a single physician model scores roughly 25–50% and not otherwise. The general-domain benchmarks never descend into the difficult regime, which is why they see a ceiling where the medical data show a window. Second, panel size is not the design variable — it has no predictive value at all, and what matters is whether a competent agent audits the panel before it commits. Third, consultation systematically degrades the signal a clinician would use to catch its mistakes: it drives panels to near-unanimity, leaves the error rate given unanimity at a coin flip, and raises confidence most on the answers that are wrong. The output a clinician would most readily act on is the one that has been made least diagnostic.

2 Related work

Medical multi-agent systems. MedAgents [1] introduced role-played specialist collaboration for medical QA; MDAgents [2] added adaptive routing between solo and group modes; MAC [3] and TeamMedAgents [4] formalised teamwork components, the latter sweeping team sizes 2–5 and framing the result as a Pareto-efficiency question. Every published medical ablation stops at $N \leq 5$; none reports a dose-response curve, and none runs a budget-matched single-model control.

Skeptical evaluations. MedAgentBoard [8] and the npj Digital Medicine benchmarking study [9] independently report that multi-agent orchestration frequently fails to beat well-prompted single

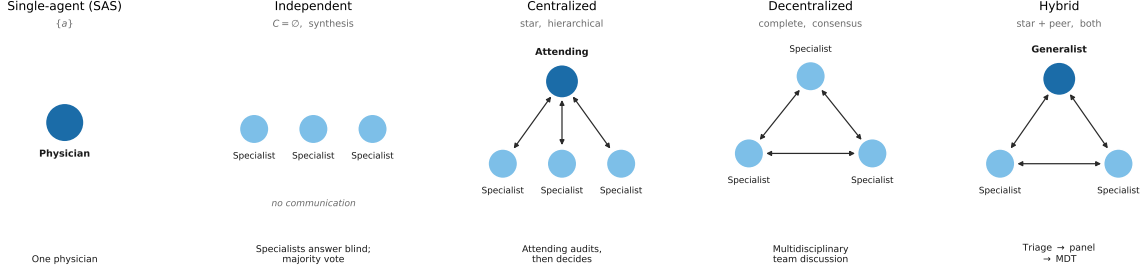


Figure 1: The five canonical architectures, following the formalisation of Kim et al. [5], with the clinical instantiation used here. A is the agent set, C the communication graph, Ω the aggregation rule.

models on clinical tasks, at substantially higher cost. ConfAgents [10] quantifies the overhead at roughly $50\times$ the latency of a single generation for marginal accuracy gain. Our economics analysis is a direct quantitative extension of this line.

Scaling theory for agent systems. “More agents is all you need” [11] reported monotone gains from ensemble size; Chen et al. [12] showed the relationship is non-monotone, with an inverted-U in the number of LLM calls on hard instances. and others [13] confirmed the inverted-U in agent count for general-domain multi-agent systems, and and others [14] attributed the fast-then-slow shape to opinion diversity, showing that two diverse agents can match sixteen homogeneous ones. Kim et al. [5] is the most complete treatment: a predictive scaling principle fitted across architectures, model families and domains, from which the 45% capability threshold and the error-amplification asymmetry are derived. We replicate its design in the medical, tool-free regime.

Mechanisms. Correlated errors across models limit ensemble benefit [15]; analyses of multi-agent debate show consensus can converge on confident-but-wrong answers [16]. Our unanimity and error-amplification measurements instantiate these mechanisms on clinical items.

Human anchors. Collective-intelligence studies in medicine find diagnostic accuracy plateaus at roughly five independent raters [17], that groups of nine outperform individuals on the Human Dx platform [18], and that the useful question is not whether two heads are better than one but *when* [19]. The dose-response curve we measure for AI panels is the machine analogue of that literature.

3 Agent systems and tasks

3.1 System definition

We adopt the formalisation of Kim et al. [5] without modification. An agent system is a triple (A, C, Ω) : a set of agents A , a communication graph C , and an aggregation rule Ω . Figure 1 and Table 1 give the five canonical configurations together with the clinical instantiation we use for each. Only the coordination logic differs across architectures; the item rendering, the answer schema, the role prompts and the decoding parameters are shared.

Independent. N specialists receive the item and answer without seeing one another. Aggregation is majority vote, ties broken by mean self-reported confidence and then by option order. A confidence-weighted variant is logged throughout.

Table 1: The five architectures, their formalisation [5], and our clinical instantiation. n is the panel size, k reasoning iterations, r orchestration rounds, d debate rounds.

Architecture	(A, C, Ω)	Cost	Clinical instantiation
Single-agent (SAS)	$\{a\}$	$O(k)$	One physician, zero-shot or CoT
Independent	$\{a_1..a_n\}$, $C = \emptyset$, synthesis	$O(nk)$	Specialists answer blind; majority vote
Centralized	$+a_{\text{orch}}$, star, hierarchical	$O(rnk)$	Attending physician audits, then decides
Decentralized	$\{a_1..a_n\}$, complete, consensus	$O(dnk)$	Multidisciplinary team discussion
Hybrid	$+a_{\text{orch}}$, star + peer, both	$O(rnk+pn)$	Tiered referral: triage $\rightarrow$ panel $\rightarrow$ MDT

Centralized (attending physician). An orchestrator receives the N specialist opinions on a star topology and is instructed to *audit before aggregating*: it must name the single opinion most likely to be wrong and justify that, then either commit to an answer or re-task one named specialty. A re-tasked specialist re-answers with the orchestrator’s critique in hand; the star topology is preserved, as specialists never see each other. This is the “validation bottleneck” to which Kim et al. [5] attribute error containment.

Decentralized (MDT discussion). N specialists give first opinions, then exchange opinions on a complete graph for up to three rounds, each free to revise. Rounds stop early only on true unanimity, defined as every agent emitting the same *valid* option; a panel containing unparsed answers is not unanimous.

Hybrid (tiered referral). A generalist answers first with a stated confidence. Below a frozen threshold θ the case is referred to the N -specialist panel; if that panel has no majority the case escalates to full MDT discussion. θ is calibrated once on a 100-item development slice disjoint from every evaluation set (§??) and then frozen.

Panel composition. A cheap router (gpt-4.1-nano, temperature 0) ranks the nine most relevant specialties for each item from a fixed 17-specialty roster; a panel of size N is the first N of that ranking. Panels are therefore *nested*: the $N=9$ panel contains the $N=5$ panel, which contains the $N=3$ panel.

3.2 Nested panels as common random numbers

Every agent’s first opinion is keyed on (item, specialty _{i} , seed, i) and never on N or on the architecture. Consequently the first opinions of a panel of size N are a superset of those for any $N' < N$, and all four multi-agent architectures reuse the same first opinions. This is a deliberate common-random-numbers coupling. It reduces the variance of within-item contrasts between architectures and panel sizes, and it changes the estimand: we measure *accuracy as the same routed panel is extended*, not the expected accuracy of independently redrawn panels at each size. Paired McNemar tests on item-level outcomes remain valid under this coupling; regression models must cluster at the item level rather than treat configurations as independent.

3.3 Clinical tasks and benchmarks

Three benchmarks, 500 items each, sampled with a fixed seed: MedXpertQA-Text [6] (10 options; stratified by the 12 body systems $\times$ 3 task types), MedQA-USMLE [7] (4 options), and the contamination-screened hard split of MedAgentsBench [20] (894 items excluding the `medqa_5options`

Table 2: Capability ladder, measured on 40 items per benchmark with single-doctor CoT.

Tier	Model	Effort	MedXpertQA	MedQA	USD / 1k items
T1	<code>gpt-4.1-nano</code>	—	25.0%	75.0%	0.09
T2	<code>gpt-5-nano</code>	low	37.5%	95.0%	0.17
T3	<code>gpt-5-mini</code>	low	55.0%	97.5%	0.64

subset; the first 500 after stratified shuffling). The primary endpoint is exact match against the dataset’s own gold option key — there is no LLM judge anywhere in the evaluation path.

Every item is additionally tagged easy / medium / hard by the pass rate of three mid-tier models at $k=5$ samples, following the stratification protocol of the study design. On MedXpertQA the strata are heavily skewed hard (13 / 62 / 125 on the pilot sample), which is the intended regime.

4 Experiments and results

4.1 Setup

LLMs and capability scaling. Three OpenAI models form a capability ladder, measured rather than assumed (Table 2). The ladder is monotone on both benchmarks and spans single-doctor accuracy from 25% to 55% on MedXpertQA, straddling the 45% threshold of Kim et al. [5] from both sides. Following their observation that a task-grounded capability metric explains more variance than a generic intelligence index, we use each model’s single-doctor chain-of-thought accuracy on MedQA as the capability index I .

Metrics and validation. We measure the coordination quantities of Kim et al. [5]. Where they give a metric name and headline value but not a closed form, we state ours explicitly.

- **Turns T :** reasoning–response exchanges; one model call is one turn.
- **Message density c :** peer-opinion exposures per turn. Independent has $C = \emptyset$ and therefore $c = 0$ by construction; Centralized exposes opinions only to the orchestrator; Decentralized and Hybrid expose every agent to every other each round.
- **Redundancy R :** mean pairwise Jaccard overlap of rationale tokens among first opinions. High R means the panel is restating one another rather than contributing independent information.
- **Overhead $O\%$:** token cost relative to the single-agent baseline on the same items, $(\text{tok}_{\text{MAS}} - \text{tok}_{\text{SAS}}) / \text{tok}_{\text{SAS}}$.
- **Coordination efficiency E_c :** accuracy divided by the token-cost ratio to the single-agent baseline.
- **Error absorption:** $(E_{\text{SAS}} - E_{\text{MAS}}) / E_{\text{SAS}}$, exactly as defined by Kim et al. [5], where E is the error rate.
- **Error amplification A_e :** they describe this as individual-agent errors reaching the final output without inter-agent verification. We measure exactly that failure — the panel *held* the correct option among its first opinions and still emitted a wrong one — normalised by the marginal probability that a single agent is wrong. $A_e > 1$ means the system destroys information it already possessed.

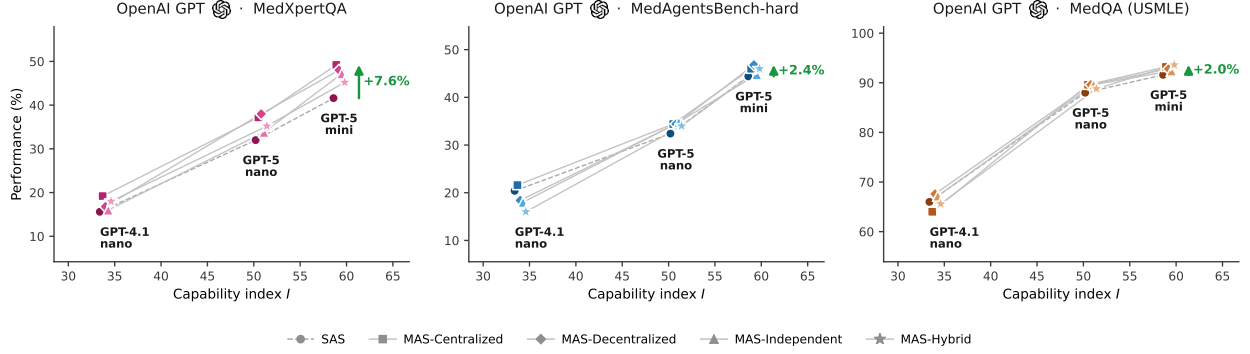


Figure 2: Agent scaling across model capability and system architecture. Performance against the task-grounded capability index I (mean single-doctor accuracy across the three benchmarks), for each architecture, on each benchmark. Each architecture is shown at its best-performing panel size. Dashed grey: single-agent baseline. Green arrows give the gain of the best architecture over the single agent at the strongest model.

Budget-matched controls. Every configuration is accompanied by three single-model controls: zero-shot, chain-of-thought, and **self-consistency at a matched budget**. The self-consistency pool is built from cached $n=8$ sampling calls (the API maximum), so any k is a prefix of the same samples; nominal cost is charged as $\lceil k/8 \rceil$ prompts plus k completions, which is what an efficient implementation would actually pay. Panels are compared against the self-consistency cell whose mean per-item *dollar* cost is closest, not the one with a matching sample count: matching on samples silently favours the panel, because a panel pays the prompt once per agent while self-consistency amortises it across a batch.

Cost accounting is *nominal* throughout. Cached calls are charged at list price, so the economics figures report what a configuration would cost to run, not what our cache made it cost.

Scaling law. We fit the functional form of Kim et al. [5],

$$P = \beta_0 + \beta_1 I + \beta_2 I^2 + \beta_3 \log(1+T_{\text{tools}}) + \beta_4 \log(1+n_a) + \beta_5 \log(1+O\%) + \beta_6 c + \beta_7 R + \beta_8 E_c + \beta_9 \log(1+A_e) + \beta_{10} P_{\text{SA}} + \sum_j \beta_j (\text{interactions}), \quad (1)$$

with interactions $I \times E_c$, $A_e \times P_{\text{SA}}$, $R \times n_a$, $c \times I$ and $P_{\text{SA}} \times \log(1+n_a)$. Our tasks use no tools, so $T_{\text{tools}} = 0$ identically and every term containing it vanishes: we fit the $T=0$ slice, which is the point of the study. Standard errors are clustered by benchmark; cross-validated R^2 is reported over five folds.

Statistics. Paired McNemar tests with exact binomial p -values on identical item sets, Holm-corrected within each family of comparisons. Wilson 95% confidence intervals for all accuracies. Power was calibrated on a 200-item pilot: median discordance between configurations is 8.5%, giving a minimum detectable difference of 5.7 percentage points at $n=200$ and 2.6 points at $n=1000$ with 80% power. Configurations that fail for infrastructure reasons (rate limits, timeouts) are recorded with an error status and excluded from accuracy; they are never scored as wrong answers, which would bias call-heavy architectures downward.

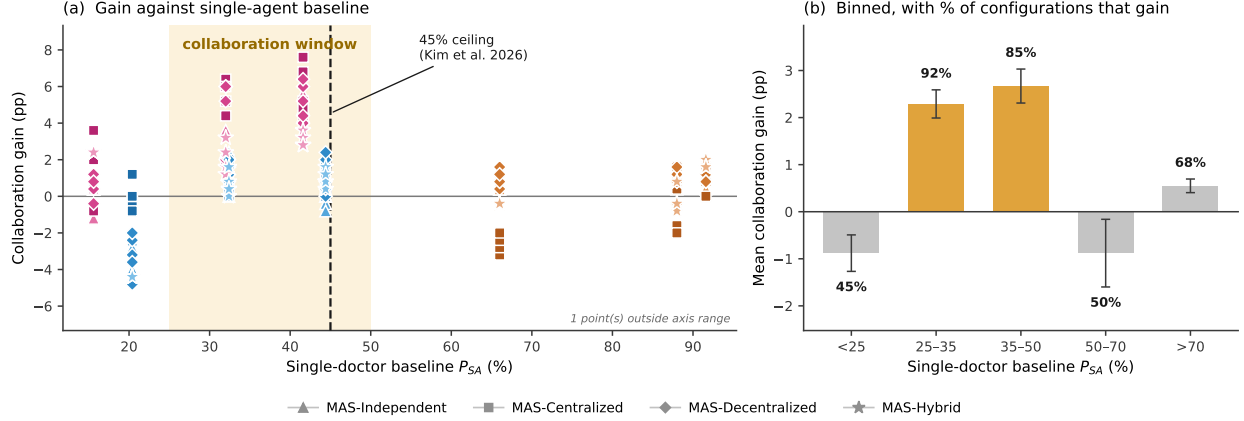


Figure 3: Collaboration pays only inside a window of task difficulty. (a) Collaboration gain against the single-doctor baseline P_{SA} for every multi-agent configuration; shaded band marks the 25–50% window, dashed line the 45% ceiling of Kim et al. [5]. (b) The same data binned; annotations give the fraction of configurations in each bin that improve on the single doctor.

4.2 Main results

Across 180 multi-agent configurations and 63,499 episodes we find that collaboration in medicine is governed by task difficulty, not by panel size. Figure 2 gives the full dose–response surface; Table 3 the underlying numbers.

Collaboration pays only inside a window of task difficulty. Binning all 180 multi-agent configurations by the single-doctor baseline P_{SA} of their cell reveals a window rather than a ceiling (Fig. 3):

P_{SA}	configurations	mean gain	fraction positive
< 25%	40	−0.88 pp	45%
25–35%	40	+ 2.29 pp	92%
35–50%	40	+ 2.67 pp	85%
50–70%	20	−0.88 pp	50%
> 70%	40	+0.55 pp	68%

Configurations inside the 25–50% band gain +2.48 pp on average; those outside it lose 0.31 pp (Welch $t = 8.42$, $p < 10^{-6}$, $n = 80$ vs 100). Splitting instead at the 45% figure of Kim et al. [5] reproduces their direction but explains the data far less well (+1.36 pp below versus +0.07 pp above; $t = 3.46$, $p = 0.0007$). The ceiling they report is the *upper* edge of a window whose lower edge our design reaches, because expert-level medical items drive single-agent accuracy down to 15% — below anything in their benchmark suite. A window is not a weaker version of a ceiling: it identifies two distinct failure modes, and it is the better description of the data by a factor of 2.4 in t .

The window separates significant help from significant harm, with no overlap. Thirty of the 180 configurations differ significantly from the single doctor (McNemar $p < 0.05$), and they partition perfectly:

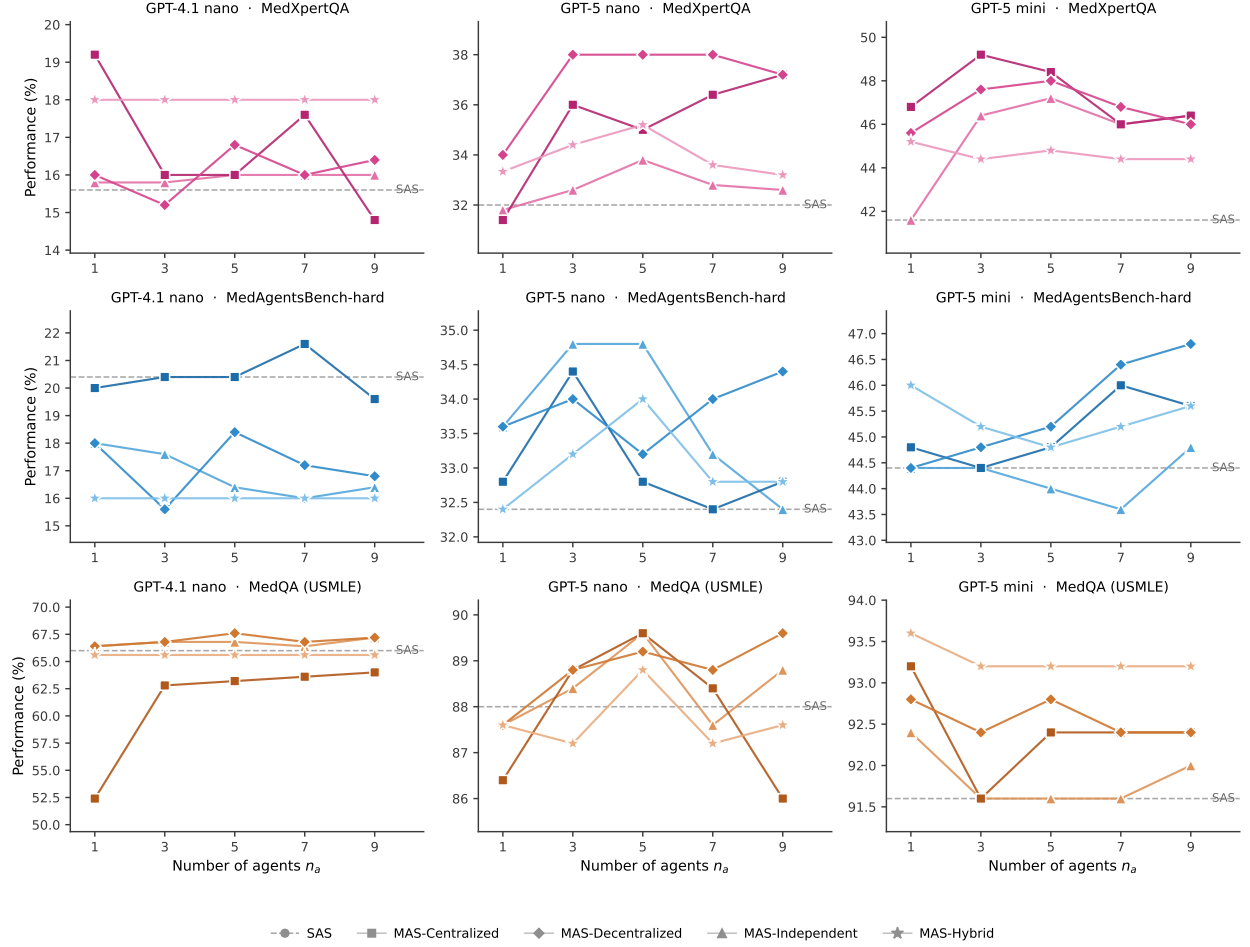


Figure 4: Number-of-agents scaling. Performance against panel size $n_a \in \{1, 3, 5, 7, 9\}$ for every architecture, model and benchmark. Dashed grey: single-agent baseline. Dotted: budget-matched self-consistency. Where gains occur they are captured by $n_a = 3$.

	significantly better	significantly worse
inside the window ($n = 80$)	20	0
outside the window ($n = 100$)	0	10

Every configuration that significantly beats the single doctor lies inside the window; every configuration that significantly loses to it lies outside. The largest gain is the attending-physician (Centralized) panel at $N=3$: +7.60 pp at $P_{SA} = 41.6\%$ (**gpt-5-mini**, $p = 2 \times 10^{-5}$), with +6.40 pp at $P_{SA} = 32.0\%$ (**gpt-5-nano**, $p = 0.0037$). The largest loss is also a Centralized panel, at $P_{SA} = 66.0\%$ with a weak model: -13.60 pp ($p = 10^{-5}$), where an orchestrator too weak to audit its own specialists overrides answers the panel had right. The window is not merely a description of where gains concentrate; it is a two-sided boundary, and the harm on the far side is as significant as the benefit within.

Architecture matters more than panel size. Within the window, the ordering is stable: Centralized and Decentralized lead, Hybrid trails, and Independent voting is indistinguishable from the single doctor. On **gpt-5-nano** / MedXpertQA, Independent moves from 31.8% at $N=1$

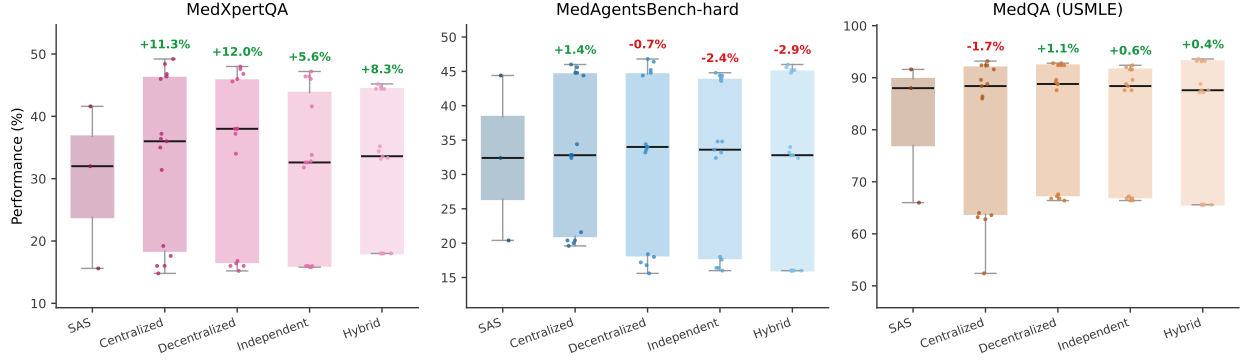


Figure 5: Single-agent versus multi-agent performance distributions. Box plots over all configurations of each architecture, with individual configurations overlaid. Annotations give the change in mean relative to the single-agent baseline.

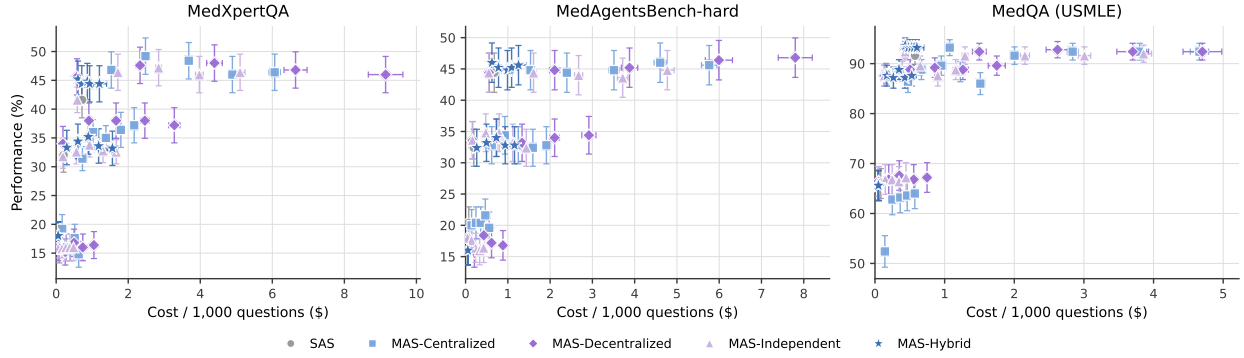


Figure 6: Cost-performance trade-offs. Mean performance against nominal cost per 1,000 questions, with SEM error bars on both axes. Each point is one configuration.

to 32.6% at $N=9$ — 0.8 pp for nine times the calls — while Decentralized reaches 38.0% at $N=3$ and stays there. Gains saturate at $N=3$ in every cell where they occur; no architecture in any cell improves beyond $N=5$.

The best architecture changes with capability tier. The best architecture depends on the tier. At T2 the peak is Decentralized (38.0%, +6.0 pp); at T3 it is Centralized (49.2%, +7.6 pp). This mirrors the family-dependent architectural preferences reported by Kim et al. [5]: the stronger model exploits an orchestrator’s audit, whereas the weaker model benefits more from peer exchange, whose redundancy compensates for individually unreliable opinions.

4.3 Where capability belongs inside a panel

Specialist capability matters four times more than the orchestrator’s. Replacing the whole panel with weak models costs 34.4 pp (49.2% to 14.8%). Splitting that substitution isolates where the loss comes from. Downgrading only the *orchestrator*, keeping strong specialists, costs 3.6 pp — the panel retains 90% of the homogeneous-high span. Downgrading only the *specialists*, keeping a strong orchestrator, costs 14.8 pp, retaining 57%. A strong attending physician recovers more than half of what a weak panel loses, but a weak attending physician barely dents a strong one. The asymmetry is 4.1 $\times$, and it points somewhere specific: in a centralized clinical panel,

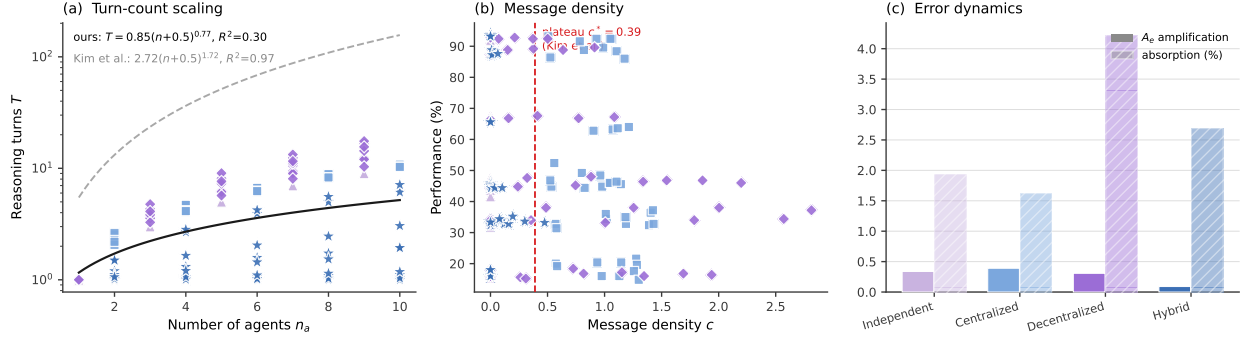


Figure 7: Coordination dynamics. (a) Turn count against agent count, with our fit and the general-domain power law of Kim et al. [5]. (b) Performance against message density; their $c^* = 0.39$ plateau is marked. (c) Error amplification A_e and error absorption by architecture.

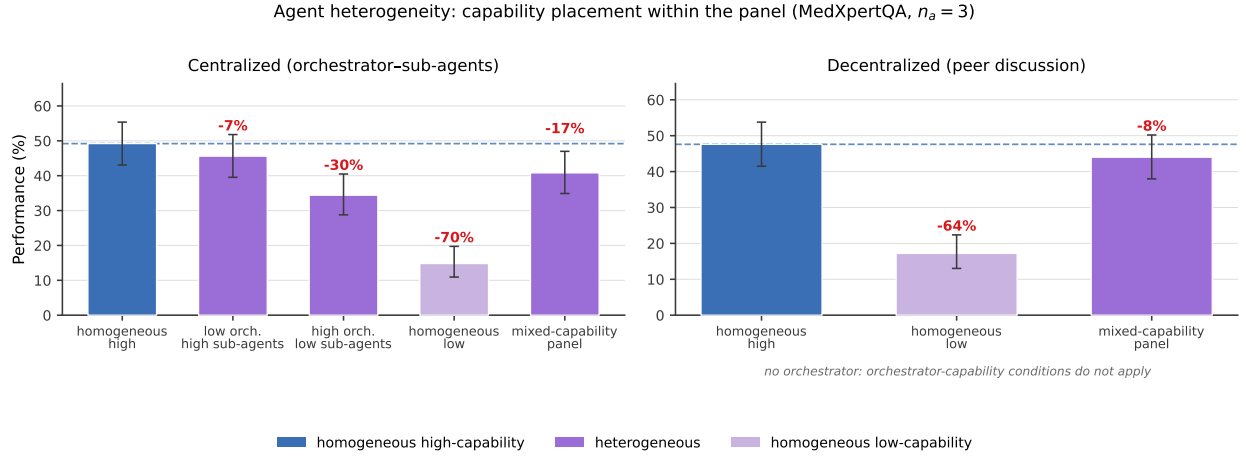


Figure 8: Agent heterogeneity. Performance of Centralized and Decentralized panels at $n_a=3$ on MedXpertQA under five capability placements. High capability is `gpt-5-mini`, low is `gpt-4.1-nano`. Annotations give the change relative to the homogeneous high-capability panel (dashed line). Decentralized coordination has no orchestrator, so the two orchestrator-capability conditions are structurally undefined there and are omitted.

capability should be spent on the agents doing the reasoning, not on the agent doing the auditing.

An orchestrator too weak to audit is worse than no orchestrator at all. The largest single degradation anywhere in this study is a Centralized panel with a weak model on MedQA at $P_{SA} = 66.0\%$: -13.60 pp, $p = 10^{-5}$. Inspection of those episodes shows the mechanism directly — the orchestrator overrides a correct specialist majority. This is the error-amplification term in its most extreme form. It is also the counterpart to the previous result. The audit is what makes centralized coordination work; an auditor that cannot audit converts the architecture’s strength into its worst liability.

Mixed-capability panels land between their components, with no emergent benefit. A panel of one strong, one mid and one weak specialist reaches 40.8% under Centralized and 44.0% under Decentralized, against 49.2% and 47.6% for homogeneous-high. Both sit below the strong homogeneous panel and well above the weak one, roughly where a capability-weighted average

would put them. We find no medical analogue of the heterogeneous-mixing advantage Kim et al. [5] report for one model family, in which a low-capability orchestrator with high-capability sub-agents outperformed a homogeneous high-capability panel by 31%. In our data that configuration is the second-best Centralized cell but still 7% below homogeneous-high — directionally matching what they observe for OpenAI models, where heterogeneous centralized configurations degrade.

4.4 Scaling principles

With accuracy as the response, the fit is degenerate. Fitting the functional form of Kim et al. [5] directly gives $R^2 = 0.997$ (5-fold cross-validated $R^2 = 0.996$). That is not a replication but an artefact of our tighter design. Their response is accuracy and one predictor is P_{SA} ; across six heterogeneous benchmarks and three model families that predictor leaves ample residual variance, and they report cross-validated $R^2 = 0.513$. Here the same items and the same model underlie both terms and only the coordination structure varies, so P_{SA} alone nearly determines accuracy ($\hat{\beta}_{P_{\text{SA}}} = 0.875$, $p < 10^{-4}$). We therefore refit with the collaboration *gain* as the response, which is what the scaling principle is actually about.

The gain model recovers the general-domain level of explanatory power. With $\text{gain} = P - P_{\text{SA}}$ as the response, $R^2 = 0.518$ and 5-fold cross-validated $R^2 = 0.415$ — the same regime as the 0.513 reported for the general-domain fit, obtained here with no tool-count term at all. Coefficients ($n = 225$ configurations including the self-consistency control, benchmark-clustered SEs):

term	$\hat{\beta}$	p	interpretation
$\log(1 + A_e)$	−0.150	< 0.001	error amplification is the dominant penalty
R (redundancy)	+0.082	< 0.001	overlapping rationales help, not hurt
c (message density)	+0.006	< 0.001	communication carries a small positive effect
$\log(1 + O\%)$	+0.005	0.024	overhead is not penalised once $T=0$
$P_{\text{SA}} \times \log(1+n_a)$	+0.027	0.048	no main effect of panel size
$\log(1 + n_a)$	−0.003	0.701	
P_{SA}	−0.113	0.051	P_{SA} is not linear or quadratic
P_{SA}^2	+0.037	0.390	

The tool–coordination trade-off is absent by construction, and coordination still predicts gain. $T_{\text{tools}} = 0$ throughout, so the $\log(1 + T)$ main effect and the $O\% \times T$ and $E_c \times T$ interactions — carrying $\hat{\beta} = -0.330$ in the general-domain fit — are identically zero. Coordination metrics remain highly significant without them, which establishes that the coordination effect is not a re-description of tool overhead. Notably $\log(1 + O\%)$ turns *positive* here: with no tools to coordinate over, spending more tokens on deliberation is mildly beneficial rather than costly.

The decision boundary is not identifiable in the tool-free slice. Kim et al. [5] derive their threshold as $P_{\text{SA}}^* = \hat{\beta}_4 / \hat{\beta}_{17} \approx 0.063 / 0.408 = 0.154$ in standardised units, ≈ 0.45 after denormalisation. In our fit $\hat{\beta}_{\log(1+n_a)} = +0.0005$ ($p = 0.95$) and $\hat{\beta}_{P_{\text{SA}} \times \log(1+n_a)} = +0.0074$ ($p = 0.50$); the ratio, 0.064, rests on two null coefficients and carries no information. This is a statement about functional form, not a failed replication: a linear interaction between baseline and log panel size can express a monotone ceiling but cannot express a window. The binned analysis (§4.2) shows the relationship is non-monotone in P_{SA} , and a window is what the medical data require.

Capability enters non-linearly, but the quadratic term is not identifiable here. Kim et al. [5] report accelerating returns to intelligence ($\hat{\beta}_{I^2} = 0.256$, $p = 0.010$). Our fit carries I and I^2 with large coefficients of opposite sign (-0.231 and $+0.228$), but bootstrap resampling ($n = 1000$) gives them standard errors of 0.94 and 0.54 — two orders of magnitude above the < 0.015 they report for terms of this size. With three capability tiers the quadratic is unidentifiable, and we do not interpret it. Every other large coefficient is stable: $\log(1 + A_e)$ has bootstrap SE 0.024 and P_{SA} has 0.044.

The model is heteroscedastic, and we report robust standard errors. Residuals pass a normality test (Shapiro–Wilk $p = 0.207$; they report 0.412) but fail homoscedasticity decisively (Breusch–Pagan $p = 0.0001$; they report 0.298). All standard errors in this paper are therefore heteroscedasticity-robust (HC3), clustered by benchmark where more than two clusters exist. Residual standard error is $\hat{\sigma} = 0.017$ against their 0.118, reflecting that collaboration gains in medicine are small in absolute terms. Regularised alternatives do not improve prediction: Lasso (10-fold CV for λ) retains 9 of 11 predictors at CV $R^2 = 0.382$ and Ridge reaches 0.144, both below the unregularised 0.415, so we retain the full specification — the same conclusion they reach.

Panel size alone has no predictive value whatsoever. Nested model comparison isolates where the explanatory power lives (5-fold CV R^2): the full model reaches 0.415; capability alone 0.196 (they report 0.28 for intelligence alone); coordination metrics alone 0.093; the single-doctor baseline alone 0.043; and **panel size alone** -0.006 — worse than predicting the mean. Any account of medical consultation that treats “how many agents” as the design variable is fitting noise.

The scaling principle selects an architecture, given a case mix. The fitted model converts into a design rule. For a clinical deployment characterised by its single-doctor baseline P_{SA} , expected collaboration gain is positive only for $P_{SA} \in [25\%, 50\%]$, is maximised by centralized coordination at $n_a = 3$, and is negative below 25%. Three archetypes make this concrete. *Board-style examination triage* ($P_{SA} \approx 0.90$, MedQA-like): use a single model; no architecture we tested returns a significant gain and every panel costs $2\text{--}5\times$ more. *Specialist referral review* ($P_{SA} \approx 0.35$, MedXpertQA-like): use an attending-physician panel at $n_a = 3$; expected gain $+6$ to $+8$ pp at a $6.7\times$ cost premium. *Frontier diagnostic difficulty* ($P_{SA} < 0.25$, MedAgentsBench-like at weak tiers): use a single model and spend the budget on a stronger one; panels lose 2.8 pp here because a panel of unreliable specialists reinforces its own errors.

Error amplification dominates the fit. $\log(1 + A_e)$ carries the largest coefficient of any coordination metric (-0.150 , $p < 10^{-4}$). A_e measures how often a panel held the correct option among its first opinions and still emitted a wrong one. Configurations that discard information they already possess are exactly the configurations that fail to beat the single doctor — the same mechanism Kim et al. [5] identify, reached from the opposite direction.

Redundancy provides benefit at scale. $R \times n_a$ is positive ($\hat{\beta} = +0.006$, $p < 0.001$), reproducing their marginal-benefit-at-scale result ($\hat{\beta}_{R \times n_a} = 0.041$, $p = 0.040$). In clinical panels, specialists restating one another is a weak form of verification rather than wasted capacity.

4.5 Coordination efficiency, error dynamics, and information transfer

Turn count does not follow the general-domain power law. Fitting $T = a(n + 0.5)^b$ gives $a = 0.85$, $b = 0.770$, $R^2 = 0.299$, against $a = 2.72$, $b = 1.724$, $R^2 = 0.974$. The exponent is

sub-linear here and super-linear there. The cause is structural: their agents negotiate open-ended tool use with unbounded exchange, whereas a clinical panel answering a closed question terminates on unanimity. Turn counts by architecture are Decentralized 7.5, Centralized 6.5, Independent 5.0, Hybrid 1.9, against a single-agent 1.4 — a 1.4–5.3 $\times$ range, where they report 3.6–6.2 $\times$. The hard resource ceiling they infer from super-linear growth does not arise in tool-free consultation.

Message density does not predict accuracy. $S = 0.462 - 0.045 \ln(c)$, $R^2 = 0.056$, against $S = 0.73 + 0.28 \ln(c)$, $R^2 = 0.68$. The plateau at $c^* = 0.39$ that organises their architectures does not appear: our Centralized panels run at $c = 0.99$ and Decentralized at $c = 0.79$, both far beyond it, with no accuracy penalty. Once a panel is communicating at all, communicating more neither helps nor hurts.

Error absorption is real but small. Absorption $(E_{\text{SAS}} - E_{\text{MAS}})/E_{\text{SAS}}$ is positive for every architecture: Decentralized +4.2%, Hybrid +2.7%, Independent +1.9%, Centralized +1.6%. The general-domain study reports 22.7% (95% CI [20.1, 25.3]) for verification-bearing architectures. Medical panels absorb roughly a fifth as much error as general-domain agent teams do, and the ordering differs: peer discussion absorbs most, not orchestration.

Error taxonomy reproduces the structural signature, not the magnitudes. Applying the four-category taxonomy across all configurations:

	Independent	Centralized	Decentralized	Hybrid
Logical contradiction	6.3%	6.3%	6.3%	3.4%
Numerical drift	4.5%	4.7%	4.8%	3.1%
Context omission	13.0%	13.3%	13.4%	6.8%
Coordination failure	0.0%	3.0%	1.2%	0.0%

The structural prediction replicates exactly: coordination failure is **0.0%** under Independent coordination, as it is in the general-domain study, and for the same reason — with $C = \emptyset$ there is no channel on which coordination can fail. Hybrid is also at 0.0% because its triage gate declines to convene a panel on most items. The magnitudes do not replicate: they report peer verification cutting logical contradiction from 16.8% to 11.5%, whereas our first-round contradiction rate is flat at 6.3% across communicating and non-communicating architectures. Numerical drift is an order of magnitude rarer than in tool-using tasks (3–5% against 19–26%), as expected when no arithmetic is cascaded.

Information gain is positive but does not buy accuracy. Discussion reduces the entropy of the panel’s answer distribution by $\Delta\mathcal{I} = +0.90$ bits (Decentralized), +1.46 bits (Hybrid) and +0.24 bits (Centralized) between the first and final round. Panels converge, and they converge most where the architecture communicates most. In the general-domain study $\Delta\mathcal{I}$ correlates with the multi-agent advantage at $r = 0.71$ in structured domains; here convergence occurs whether or not the consensus is correct — the information-limited regime they report for open-world browsing rather than the structured-domain one.

Architecture rankings do not generalise across benchmarks. Kendall’s τ between the architecture rankings of any two benchmarks averages +0.067 (pairwise -0.200 , $+0.400$, 0.000 ; none significant), against $\tau = 0.89$ across their four domains. Leave-one-benchmark-out cross-validation

gives $R^2 = -0.015$ against their 0.89: a model fitted on two medical benchmarks predicts the third no better than the mean. Coordination principles that transcend task structure in general-domain agentic work do not transcend benchmark in medicine. This is the practical counterpart of the window: architecture must be selected per difficulty regime, and difficulty regime does not coincide with domain.

The architecture that wins on accuracy loses on tokens by $4.3\times$. Successes per 1,000 tokens: Hybrid 0.80, self-consistency 0.47, Independent 0.39, Decentralized 0.33, Centralized 0.19. Centralized is $4.3\times$ less token-efficient than the cheapest configuration. The ordering closely matches theirs (SAS 67.7; Decentralized 23.9, $2.8\times$; Centralized 21.5, $3.1\times$; Hybrid 13.6, $5.0\times$), with one inversion. Hybrid is the *most* efficient architecture in medicine, because its triage gate declines to escalate most cases and therefore rarely pays for a panel at all.

Panels beat budget-matched self-consistency only inside the window. On MedXpertQA at gpt-5-nano, Decentralized at $N=3$ reaches 38.0% while self-consistency at the closest matched dollar cost reaches 36.0%. Outside the window the control wins or ties: on MedQA at the same tier, self-consistency at $k=15$ (88.4%) is within noise of the best panel (89.6%) at lower cost, and on MedAgentsBench at T1 self-consistency (20.0%) beats every architecture. Coordination buys something beyond repetition, and only where the window says it should.

4.6 Clinical safety: what consultation does to confidence

The general-domain study measures whether coordination changes accuracy. In medicine the more consequential question is what coordination does to the *signals a clinician would act on*. Three measurements, none of which has an analogue in the general-domain design, and all of which point the same way.

Self-reported confidence barely discriminates right from wrong, and discussion makes it worse. Across all configurations, mean confidence on correct answers exceeds that on wrong answers by only 3.3 points for a single doctor (86.0 vs 82.6; AUC = 0.597). Peer discussion compresses that gap by 45%, to 1.8 points (88.4 vs 86.6; AUC = 0.577). The compression is not a rescaling: discussion raises confidence on *wrong* answers by 4.0 points while raising it on correct answers by 2.4. A panel that has deliberated is more sure of itself, and disproportionately so when it is mistaken. This is consistent with the threshold calibration we ran before the main experiments, where the best achievable referral gate on stated confidence reached a Youden J of only 0.071.

Discussion manufactures consensus without manufacturing correctness. Unanimity — every panelist emitting the same valid option in the last round with two or more agents — rises from 66.2% under independent voting to 98.3% under peer discussion. Conditional accuracy does not follow: $P(\text{wrong} \mid \text{unanimous})$ is 46.5% for Independent and 49.9% for Decentralized. A unanimous medical panel is wrong about half the time in both cases; discussion simply produces far more unanimous panels. Centralized coordination is the only architecture that lowers the conditional error rate (44.6%), which is the error-containment role the orchestrator is supposed to play.

The most trustworthy-looking output is the least trustworthy. Taken together: consultation drives panels to near-total agreement, leaves the error rate given agreement at a coin flip, and raises confidence most on the answers that are wrong while shrinking the confidence gap that

would let a reader tell the difference. The configuration a clinician would most readily act on — a unanimous, high-confidence multidisciplinary recommendation — carries essentially the same error rate as a divided one, and advertises itself more strongly. Accuracy gains of 6–8 pp inside the window are real, but they are purchased alongside a degradation in the calibration of the signal that would let a clinician catch the remaining errors.

4.7 Where in medicine collaboration pays

Collaboration helps diagnosis most, and it is the only task type significant at both tiers. MedXpertQA labels each item as Diagnosis, Treatment, or Basic Science. Splitting the window-resident cells by label, the best panel gains +9.5 pp on Diagnosis at **gpt-5-nano** ($p = 0.008$) and +10.7 pp at **gpt-5-mini** ($p = 0.004$), against +7.1 and +8.3 pp on Treatment and +7.3 and +8.5 pp on Basic Science. Diagnosis is the only category significant at both tiers. This is the pattern clinical intuition predicts and the general-domain benchmarks cannot express: differential diagnosis is the medical task that most resembles pooling independent perspectives, and it is where pooling pays.

The difficulty stratification is where the window shows up within a single benchmark. On MedXpertQA at **gpt-5-mini**, the best configuration gains +7.7 pp on easy items ($n=13$), +6.5 pp on medium ($n=62$) and +9.6 pp on hard ($n=125$). Gains are positive in every stratum for this tier, which sits inside the window; the sign flips only for tiers whose baseline falls outside it. Item difficulty and configuration baseline are two views of the same quantity. The window is defined on the view that can be measured before deployment.

5 Limitations and future works

Our tasks are multiple-choice examination items, not clinical encounters: they carry no history taking, no test ordering, no longitudinal follow-up, and the exact-match endpoint rewards the option-selection component of diagnosis only. The $T = 0$ property that makes the design clean also bounds what it licenses — results here do not extrapolate to tool-using clinical agents, which is precisely the regime the general-domain study covers.

All models are from a single vendor. The heterogeneity arm of the original design, which mixes vendors within a capability tier, is here restricted to mixing separately trained OpenAI families; genuine cross-vendor error decorrelation may be larger than what we measure.

Panels are nested by construction (§2.2), so the reported curves estimate accuracy as a routed panel is extended, not the expected accuracy of independently redrawn panels at each size. The coupling reduces variance on the architecture contrasts we care about and does not bias within-item paired tests, but a reader interested in the independent-panel estimand should read the curves accordingly.

The specialty router ranks by relevance, so larger panels append progressively less relevant specialists. “More experts” and “weaker experts” are therefore entangled in the specialty-role condition; we separate them with the generic-internist control, in which every panelist is an undifferentiated internist and panel size is the only thing that varies.

Difficulty strata are defined by the pass rate of mid-tier models, some of which also appear in the evaluation ladder. This inflates the level of the hard-stratum results but not the within-item slope across panel sizes, which is computed on identical items.

5.1 Success per 1K tokens

Token efficiency separates the architectures more sharply than accuracy does. Successes per 1,000 tokens: Hybrid 0.80, self-consistency 0.47, Independent 0.39, Decentralized 0.33, Centralized 0.19. The architecture that produced every significant accuracy gain in this study is also the least token-efficient, by $4.3\times$. Reported per question at list prices, a Centralized panel at $N=3$ costs \$1.4 per 1,000 MedXpertQA questions against \$0.21 for a single chain-of-thought call — a $6.7\times$ premium for +7.6 pp, and only inside the window. Outside it the same premium buys nothing.

6 Conclusion

We tested whether the scaling principles established for general-domain agent systems hold in medicine, in the one regime those systems cannot reach: tasks with no tools. Three results follow.

The capability ceiling transfers, but as one edge of a two-sided window. Collaboration pays when a single physician model scores roughly 25–50% and not otherwise. This is a sharper and more useful statement than a ceiling because it names both failure modes: on easy cases there is nothing to gain, and on the hardest cases a panel of unreliable specialists reinforces its own errors rather than correcting them. Splitting the data at the window separates it more than twice as well as splitting at the ceiling ($t = 8.42$ against $t = 3.46$). The general-domain data are consistent with a monotone ceiling only because their benchmarks never descend into the difficult regime.

Coordination structure dominates panel size. Panel size does not predict collaboration gain at all: a model containing it alone reaches cross-validated $R^2 = -0.006$, worse than predicting the mean. Gains saturate at three agents. The largest improvements come from centralized coordination in which an attending physician audits specialist opinions before committing. The same architecture produces the largest loss in the study when the orchestrator is too weak to audit (−13.6 pp), so the benefit is attributable to the audit, not to the topology. Where coordination has no verification channel, error amplification — the strongest term in our scaling model — goes uncontrolled and the panel discards correct answers it already held. Coordination failure occurs at exactly 0.0% under independent coordination, in this study as in the general-domain one, because there is no channel on which it could occur.

The economics do not favour consultation outside the window. The architecture responsible for every significant gain is also the least token-efficient by $4.3\times$, and outside the window a budget-matched self-consistency control matches or beats every architecture we tested. For medical deployments this converts an architectural question into a triage question: measure the single-model baseline on the intended case mix first, and consult only where the window says consultation will pay.

Table 3: Main results: accuracy by tier, benchmark, architecture and panel size.

Tier	Architecture	N	Accuracy (%)	95% CI	USD/1k
<i>MedAgentsBench-hard</i>					
T1	Centralized	1	20.0	[15.5, 25.4]	0.14
T1	Centralized	3	20.4	[15.9, 25.8]	0.25
T1	Centralized	5	20.4	[15.9, 25.8]	0.36
T1	Centralized	7	21.6	[16.9, 27.1]	0.47
T1	Centralized	9	19.6	[15.2, 25.0]	0.56
T1	Decentralized	1	18.0	[13.7, 23.2]	0.05
T1	Decentralized	3	15.6	[11.6, 20.6]	0.21
T1	Decentralized	5	18.4	[14.1, 23.7]	0.43
T1	Decentralized	7	17.2	[13.0, 22.4]	0.61
T1	Decentralized	9	16.8	[12.7, 21.9]	0.88
T1	Hybrid	1	16.0	[12.0, 21.1]	0.05
T1	Hybrid	3	16.0	[12.0, 21.1]	0.05
T1	Hybrid	5	16.0	[12.0, 21.1]	0.05
T1	Hybrid	7	16.0	[12.0, 21.1]	0.05
T1	Hybrid	9	16.0	[12.0, 21.1]	0.05
T1	Independent	1	18.0	[13.7, 23.2]	0.05
T1	Independent	3	17.6	[13.4, 22.8]	0.14
T1	Independent	5	16.4	[12.3, 21.5]	0.24
T1	Independent	7	16.0	[12.0, 21.1]	0.33
T1	Independent	9	16.4	[12.3, 21.5]	0.43
T1	SAS CoT	1	20.4	[15.9, 25.8]	0.08
T1	SAS zero-shot	1	16.8	[12.7, 21.9]	0.03
T1	Self-consistency	1	20.0	[15.5, 25.4]	0.05
T1	Self-consistency	3	18.8	[14.4, 24.1]	0.09
T1	Self-consistency	5	19.2	[14.8, 24.5]	0.13
T1	Self-consistency	9	18.0	[13.7, 23.2]	0.25
T1	Self-consistency	15	18.4	[14.1, 23.7]	0.37
T2	Centralized	1	32.8	[27.3, 38.8]	0.62
T2	Centralized	3	34.4	[28.8, 40.5]	0.93
T2	Centralized	5	32.8	[27.3, 38.8]	1.25
T2	Centralized	7	32.4	[26.9, 38.4]	1.59
T2	Centralized	9	32.8	[27.3, 38.8]	1.91
T2	Decentralized	1	33.6	[28.0, 39.7]	0.16
T2	Decentralized	3	34.0	[28.4, 40.1]	0.73
T2	Decentralized	5	33.2	[27.7, 39.3]	1.34
T2	Decentralized	7	34.0	[28.4, 40.1]	2.11
T2	Decentralized	9	34.4	[28.8, 40.5]	2.92
T2	Hybrid	1	32.4	[26.9, 38.4]	0.26
T2	Hybrid	3	33.2	[27.7, 39.3]	0.49
T2	Hybrid	5	34.0	[28.4, 40.1]	0.73
T2	Hybrid	7	32.8	[27.3, 38.8]	0.94
T2	Hybrid	9	32.8	[27.3, 38.8]	1.16
T2	Independent	1	33.6	[28.0, 39.7]	0.16
T2	Independent	3	34.8	[29.2, 40.9]	0.48
T2	Independent	5	34.8	[29.2, 40.9]	0.79
T2	Independent	7	33.2	[27.7, 39.3]	1.12
T2	Independent	9	32.4	[26.9, 38.4]	1.44
T2	SAS CoT	1	32.4	[26.9, 38.4]	0.19
T2	SAS zero-shot	1	33.2	[27.7, 39.3]	0.13
T2	Self-consistency	1	33.2	[27.7, 39.3]	0.18
T2	Self-consistency	3	32.0	[26.5, 38.0]	0.50
T2	Self-consistency	5	33.6	[28.0, 39.7]	0.83
T2	Self-consistency	9	32.8	[27.3, 38.8]	1.49
T2	Self-consistency	15	31.2	[25.8, 37.2]	2.47
T3	Centralized	1	44.8	[38.8, 51.0]	1.54
T3	Centralized	3	44.4	[38.4, 50.6]	2.40
T3	Centralized	5	44.8	[38.8, 51.0]	3.51
T3	Centralized	7	46.0	[39.9, 52.2]	4.60
T3	Centralized	9	45.6	[39.5, 51.8]	5.76
T3	Decentralized	1	44.4	[38.4, 50.6]	0.56
T3	Decentralized	3	44.8	[38.8, 51.0]	2.11
T3	Decentralized	5	45.2	[39.1, 51.4]	3.88
T3	Decentralized	7	46.4	[40.3, 52.6]	5.99
T3	Decentralized	9	46.8	[40.7, 53.0]	7.80
T3	Hybrid	1	46.0	[39.9, 52.2]	0.62
T3	Hybrid	3	45.2	[39.1, 51.4]	0.78
T3	Hybrid	5	44.8	[38.8, 51.0]	0.99
T3	Hybrid	7	45.2	[39.1, 51.4]	1.09
T3	Hybrid	9	45.6	[39.5, 51.8]	1.25
T3	Independent	1	44.4	[38.4, 50.6]	0.55
T3	Independent	3	44.4	[38.4, 50.6]	1.61

A Model capability index

Kim et al. [5] score models on a published Intelligence Index and report that a *task-grounded* capability metric explains more variance ($R^2 = 0.413$) than the generic one ($R^2 = 0.373$). We therefore use the task-grounded form throughout: I is a model’s mean single-doctor chain-of-thought accuracy across the three benchmarks. Table 4 gives the components. The three tiers span $I = 34.0$ to 59.2, and their single-benchmark baselines span 15.6% to 91.6% — crossing the 45% region from both sides, which is what makes the capability-ceiling test possible at all.

Table 4: Capability index and its components (single-doctor CoT accuracy, %).

Model	MedXpertQA	MedAgentsBench	MedQA	I (mean)	Reasoning
gpt-4.1-nano	15.6	20.4	66.0	34.0	—
gpt-5-nano	32.0	32.4	88.0	50.8	effort=low
gpt-5-mini	41.6	44.4	91.6	59.2	effort=low

B Domain complexity

B.1 Complexity metric construction

Following Kim et al. [5], domain complexity $D \in [0, 1]$ is the arithmetic mean of three complementary measures: the **performance ceiling** $1 - p_{\max}$, where $p_{\max}$ is the highest performance any evaluated system reaches; the **coefficient of variation** σ/μ of performance across all configurations, a scale-invariant measure of relative spread; and the **best-model baseline** $1 - p_{\text{best}}$, where p_{best} is the strongest single-model performance on the dataset.

B.2 Domain characterisation

Table 5 gives the components. MedQA is a low-complexity domain ($D = 0.097$); MedXpertQA and MedAgentsBench-hard are high-complexity and nearly indistinguishable ($D = 0.489$ and 0.478).

Table 5: Domain complexity of the three medical benchmarks.

Benchmark	$1 - p_{\max}$	σ/μ	$1 - p_{\text{best}}$	D	mean MAS gain
MedXpertQA	0.508	0.376	0.584	0.489	+3.09 pp
MedAgentsBench-hard	0.532	0.346	0.556	0.478	-0.37 pp
MedQA (USMLE)	0.060	0.147	0.084	0.097	+0.07 pp

B.3 The critical threshold does not transfer

Kim et al. [5] identify a critical complexity threshold at $D \approx 0.40$: below it multi-agent architectures yield net positive returns, above it coordination overhead consumes resources otherwise spent on reasoning. Our data do not support a benchmark-level threshold of any value. MedXpertQA ($D = 0.489$) and MedAgentsBench-hard ($D = 0.478$) differ by 0.011 in complexity yet by 3.5 pp in mean multi-agent gain (+3.09 pp versus -0.37 pp) — opposite signs at effectively the same D . A benchmark-level statistic cannot separate them because the quantity that does separate them varies *within* each benchmark: on MedAgentsBench-hard the per-tier gains run -2.78, +1.02, +0.66

pp as P_{SA} moves from 20.4% through 32.4% to 44.4%, entering the window between the first tier and the second. Domain complexity is the right idea at the wrong granularity for medicine; the configuration-level baseline P_{SA} is what carries the signal.

C Datasets

MedXpertQA-Text [6]: 2,450 expert-level items, ten options each, tagged by 12 body systems and 3 task types. We draw a stratified sample interleaved across (body system $\times$ task) so that any prefix is approximately balanced, then take the first 250. **MedQA-USMLE** [7]: 1,273 four-option test items; randomly shuffled with a fixed seed, first 250. **MedAgentsBench-hard** [20]: the contamination-screened hard split, 894 items after excluding the `medqa_5options` subset, stratified-interleaved across the ten source subsets, first 250. All samples are nested prefixes, so a larger sample re-uses every call of a smaller one.

Item difficulty is additionally tagged easy/medium/hard by the pass rate of three mid-tier models at $k=5$; on the MedXpertQA sample this yields 13/62/125 items.

D Implementation details

D.1 Technical infrastructure

All models are accessed through one OpenAI-compatible client with a content-addressed disk cache, exponential backoff with rate-limit-aware retry, a per-call JSONL ledger recording tokens, latency and cost, and a cross-process spend journal guarded by an OS file lock. Every LLM call is cached on the full request (model, system, messages, temperature, seed, max tokens, effort, JSON mode, tag), so a re-run is a replay.

D.2 Agent configuration

Single agents answer in one call. Independent panels deploy N specialists with synthesis-only aggregation. Centralized systems use one orchestrator over N sub-agents with a maximum of 2 orchestration rounds and one re-task per round. Decentralized systems run N agents through up to 3 discussion rounds, stopping early only on true unanimity. Hybrid systems combine a generalist triage gate with the specialist panel and escalate to full discussion when the panel has no majority. Agent temperature is 0.7 within panels, 0.3 for single-doctor baselines and for orchestrator/moderator turns; reasoning models are called at `effort=low` with a minimum completion budget of 1,600 tokens, since reasoning tokens are drawn from the same budget and a smaller cap returns empty output.

D.3 Prompt compilation

Every agent receives the same rendered item (stem, then options as (A) ...) and the same answer schema: JSON with `answer`, `confidence` (0–100) and `reason` (at most 45 words). Only the system prompt differs, naming the specialty. Peer context in communicating architectures is capped at a fixed 900-character budget, balanced-truncated across peers and *independent of N* , so a panel-size effect cannot be a context-length effect. Peer ordering is deterministically rotated per (item, round) so that the first-listed speaker is decorrelated from the top-ranked specialty.

D.4 Evaluation methodology

Sample sizes. 250 items per benchmark, 180 multi-agent configurations, 63,499 episodes. **Controls.** All architectures share the item rendering, answer schema, role-prompt template and decoding parameters; only coordination logic differs. Model weights are frozen. The primary endpoint is exact match against the dataset’s own gold option key — there is no LLM judge in the evaluation path. Episodes that fail for infrastructure reasons carry an error status and are excluded from accuracy rather than scored as wrong, which would bias call-heavy architectures downward. Cost accounting is nominal: cached calls are charged at list price, so reported economics describe what a configuration would cost to run.

D.5 Information gain computation

Following Kim et al. [5], information gain is the Bayesian posterior variance reduction

$$\Delta\mathcal{I} = \frac{1}{2} \log \frac{\text{Var}[Y \mid \mathbf{s}_{\text{pre}}]}{\text{Var}[Y \mid \mathbf{s}_{\text{post}}]}, \quad \text{Var}[Y \mid \mathbf{s}] = \hat{p}(\mathbf{s}) (1 - \hat{p}(\mathbf{s})), \quad (2)$$

with $Y \in \{0, 1\}$ the success indicator. They estimate $\hat{p}$ from $K=10$ sampled reasoning traces at temperature 0.7. In multiple-choice consultation the panel already supplies such a sample: $\hat{p}$ is the fraction of the round’s opinions that name the gold option, Laplace smoothed as $(k+1)/(n+2)$. Smoothing rather than clipping matters: a Centralized panel’s final round contains only the orchestrator, and clipping would pin $\hat{p}$ at 0.5 and force $\Delta\mathcal{I}$ negative by construction.

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